DC Charging (CCS2)-II

Introduction

As electric vehicles (EVs) become increasingly prevalent, understanding the intricate processes behind their charging is crucial. DC fast charging, particularly using the Combined Charging System 2 (CCS2) standard, represents a significant leap in efficiency and convenience. This lecture delves into the detailed sequence of events that unfold during a DC CCS2 charging session, from the moment a charger is plugged in to the final disconnection. It's not just about transferring power; it's a sophisticated dance of communication, safety checks, and precise control between the electric vehicle and the charging station. Grasping this sequence is fundamental for anyone involved in the design, operation, or maintenance of EV charging infrastructure, ensuring safe, reliable, and efficient energy transfer to power our sustainable future.

The complexity of DC charging stems from the high power levels involved and the need for continuous, reliable information exchange. Unlike simpler AC charging, DC charging requires a dedicated communication channel to manage the flow of data between the vehicle and the supply equipment. This high-level communication (HLC) ensures that both the EV and the Electric Vehicle Supply Equipment (EVSE) are synchronized, compatible, and operating within safe parameters throughout the entire charging process. This lecture outlines the standardized steps, as detailed in international and national regulations, that govern this critical interaction, providing a comprehensive overview of the DC CCS2 charging sequence.

Core Concepts

Term	Definition	Significance
`PLC Communication`	Power Line Communication, a method where data is transmitted over existing power lines.	Enables high-level information exchange between EV and EVSE over the Control Pilot (CP) line, crucial for DC charging.
`High Level Communication (HLC)`	Communication protocol structured in seven OSI layers, used for extensive data exchange.	Facilitates detailed handshaking and parameter sharing, ensuring compatibility and safe operation during DC charging.

Term	Definition	Significance
`Control Pilot (CP) Line`	A dedicated pin in the charging connector used for communication and control signals.	Carries both low-level PWM signals and high-frequency PLC signals, acting as the primary communication channel.
`Proximity Pilot (PP) Pin`	A pin in the charging connector that detects the presence of a valid connector.	Immobilizes the vehicle upon connection, enhancing safety by preventing movement during charging.
`EVSE`	Electric Vehicle Supply Equipment, commonly known as the charging station.	The infrastructure component that supplies electrical energy to charge the EV.
`Disconnecting Device`	Internal contactors or switches within the EV and EVSE that connect/ disconnect the high-voltage circuit.	Crucial for safety, ensuring power flow is controlled and isolated when necessary, especially during pre-charge and shutdown.
`Pre-charging`	An initial phase where the EVSE gradually raises its output voltage to match the EV battery voltage.	Prevents large inrush currents and protects the battery and charging system from damage.
`OSI Layers`	Open Systems Interconnection model, a conceptual framework for communication protocols.	Provides a structured approach for the complex high-level communication in DC CCS2 charging.
`Isolation Check`	A safety procedure to ensure sufficient electrical isolation between high-voltage components and ground.	Prevents electrical hazards and ensures the integrity of the charging system before and during power transfer.

Term	Definition	Significance
`State A, B1, B2, C2/D`	Defined states of the Control Pilot (CP) line voltage and PWM duty cycle, indicating charging progress.	These states signal the current phase of the charging sequence, from unmated to ready for charging.

Detailed Analysis

The DC CCS2 charging sequence is a meticulously orchestrated process governed by standards like IEC 61851-23 and AIS 138. It involves continuous communication and safety checks between the Electric Vehicle (EV) and the Electric Vehicle Supply Equipment (EVSE).

1. Initial State and Connection (Mating)

The charging process begins with the charger plug being unmated. In this initial state, the Control Pilot (CP) line is at a voltage level of +12V, signifying "State A."

- **Mating (t0):** The moment the charger plug is inserted into the EV's inlet connector, the system transitions.
- The Proximity Pilot (PP) pin in the vehicle detects the valid connector.
- This detection instantaneously shifts the charging state to "B1."
- The PP pin signals the vehicle's Electronic Control Unit (ECU), leading to the immobilization of the vehicle. This means the high-voltage battery is disconnected from the propulsion system, a critical safety measure.

2. Initialization and Handshaking (t0 - t1)

After mating, the EVSE initiates communication and prepares for charging.

- **PWM Activation:** The EVSE changes its internal switch from a fixed 12V to an oscillator, activating Pulse Width Modulation (PWM) on the CP line. For DC charging, the duty ratio is maintained between 3% and 7%.
- **State B2:** With PWM activated, the vehicle enters "State B2." The PWM pulses now have a voltage level between +9V and -12V.
- **High-Level Communication (HLC) Start:** This is where the sophisticated communication begins. High-frequency signals are superimposed on the 1 kHz PWM signal, enabling Power Line Communication (PLC) over the CP line.

- HLC follows a seven-layer OSI protocol, facilitating the exchange of charging parameters.
- **Handshaking:** The EV and EVSE exchange information to establish compatibility.
- **EV Sends Limits:** The EV, acting as the master in communication, sends its maximum DC supply voltage and current requirements to the EVSE.
- **Safety Checks:** During this phase, the system monitors the DC voltage between DC+ and DC-. If it exceeds 60V or if incompatibility between the EV and DC supply is detected, the system initiates a shutdown.
- **Connector Locking:** Once compatibility is confirmed, the EV locks the charger plug into its inlet connector. This prevents accidental disconnection during charging. The EVSE then responds with its maximum supported voltage and current values.

3. Cable and Isolation Check (t1 - t4)

Before power transfer, thorough safety checks are performed on the cable and the high-voltage system.

- **EV Ready (t2):** The EV changes its CP state from B to C2 or D (depending on ventilation requirements), indicating that the EV is now "ready." This is reflected by the positive voltage level of the PWM going from +9V to +6V (for C2) or +3V (for D).
- **Isolation Check Request:** The EV requests a cable and insulation check after the connector lock is confirmed.
- **DC Supply Isolation Check:** The DC EVSE performs an internal isolation check, ensuring sufficient isolation between DC+ and DC-. It continuously reports the isolation state to the EV.
- The isolation resistance must be above 100 kΩ.
- **DC Supply Ready (t4):** After a successful isolation check, the DC supply indicates its status as "Valid" and changes to "Ready" with a cable check response. At this point, both the EV and the DC supply are ready for pre-charging.

4. Pre-charging Stage (t5 - t8)

Pre-charging is a critical step to safely match voltages before full power transfer.

- **EV Pre-Charge Request (t5):** The EV sends a pre-charge request to the EVSE. This request includes the maximum allowed DC current (limited to 2A to prevent inrush current) and the requested DC voltage.

- **DC Supply Response (t6):** The DC supply adapts its output to meet the EV's request.
- It limits the current to a maximum of 2A.
- The DC output voltage reaches the requested voltage within specified tolerances by time t6.
- **Disconnecting Device Closure (t7):** The EV continuously monitors the voltage deviation between the EVSE's output and its own battery voltage.
- The EV's internal disconnecting device (contactors) will only close if this deviation is less than 20V. This ensures a smooth connection and prevents high current surges.
- Once closed, current starts flowing, still limited to 2A.
- **Power Delivery Request (t8):** The EV sends a power delivery request to remove the 2A current limit, signaling readiness for full energy transfer. The DC supply provides feedback that it is ready.

5. Energy Transfer (Main Charging) (t8 - t10)

This is the primary phase where the battery is charged.

- **DC Current Request (t8):** The EV sets the DC current request to initiate the energy transfer phase.
- **DC Supply Adaptation (t8 - t9):** The DC supply adapts its output current and voltage to the requested values from the EV. It continuously responds with its current and voltage limits and status.
- **Current Reaches Request (t9):** The DC output current reaches the requested DC current value, with a slight delay.
- **Continuous Monitoring (t9 onwards):** From this point, the EV continuously initiates message cycles, requesting voltage and current. The DC supply responds by adjusting its output and reporting its status and limits.
- Continuous monitoring of the lock, isolation resistance (>100 kΩ), output voltage, current flow, and contact temperature is performed.
- **SOC Reached (t10):** This phase continues until the EV battery reaches its desired State of Charge (SOC).

6. Power Down and Disconnection (t10 - t17)

Once the target SOC is reached, the system safely terminates the charging session.

- **EV Reduces Current Request (t10 - t11):** The EV reduces its current request to complete the energy transfer. The DC supply follows this request, reducing its output current to less than 1A before disabling its output.
- **EV Disables Output Request (t11):** Once the current drops below 1A, the EV sends a message to the DC supply to disable its power output.
- **EV Opens Disconnecting Device (t11):** The EV is the first to open its internal disconnecting device (contactors), disengaging itself from the DC EVSE. This happens only after the current is below 1A.
- **DC Supply Disables Output (t12):** Upon receiving the request and observing the EV's disconnection, the DC supply disables its output and opens its contactors. It ensures no current flows towards the EV during this discharge phase.
- **DC Supply Not Ready (t13):** The DC supply reports its status as "Not Ready," indicating its output is disabled.
- **EV Changes CP State (t14):** The EV changes the state of its S2 switch, disconnecting the 1300 ohm resistance, and the CP state returns to "B" from C2 or D. This happens after receiving the "Not Ready" message or a timeout.
- The vehicle may also perform a welded contactor check to ensure contactors are not stuck.
- **EV Isolation Monitoring (t15):** The EV starts its own isolation monitoring.
- **EV Unlocks Connector (t16):** The EV unlocks the connector. This crucial step only occurs if the DC output voltage has dropped below 60V, ensuring safety during physical disconnection.
- Simultaneously, a "Session Stop Request" is sent to terminate the digital PLC-based communication.
- **Unmated (t17):** The charger plug is now free to be removed from the vehicle inlet connector.
- The EV and supply are unmated.
- The supply disables its DC output, and the lock is disabled.
- Disconnecting the vehicle connector returns the CP state to "State A."
- The PLC communication is terminated.

Key Takeaways

- [ ] DC CCS2 charging involves a sophisticated high-level communication (HLC) over the Control Pilot (CP) line using Power Line Communication (PLC) and a seven-layer OSI protocol.
- [ ] The charging sequence is standardized by IEC 61851-23 and AIS 138, ensuring interoperability and safety.
- [ ] Safety is paramount, with the Proximity Pilot (PP) pin immobilizing the vehicle upon connection and continuous isolation checks performed by both EV and EVSE.
- [ ] The EV acts as the master in communication, initiating requests for charging parameters and controlling the sequence.
- [ ] Pre-charging is a critical phase where the DC supply output voltage is matched to the EV battery voltage, with current limited to 2A, to prevent damaging inrush currents.
- [ ] The EV's internal disconnecting device only closes if the voltage deviation between the EVSE output and the EV battery is less than 20V.
- [ ] During power down, the EV first reduces current, then opens its disconnecting device, followed by the DC supply disabling its output and opening contactors.
- [ ] The connector is only unlocked by the EV after the DC output voltage drops below 60V, ensuring safe physical disconnection.
- [ ] The entire process, from mating to unmating, involves distinct CP states (A, B1, B2, C2/D) that signal the current phase of the charging sequence.

Conclusion

The detailed DC CCS2 charging sequence highlights the intricate interplay between an electric vehicle and its charging infrastructure. Far from a simple plug-and-play operation, it's a complex, multi-stage process built on robust communication protocols, stringent safety checks, and precise power management. Understanding these steps, from the initial handshaking and compatibility verification to the careful pre-charging and controlled power-down, is essential for appreciating the engineering behind modern EV charging. This systematic approach ensures not only the efficient transfer of energy but also the longevity of vehicle batteries and the safety of users.

As we look ahead, the reliability and intelligence embedded in standards like CCS2 will continue to evolve, supporting faster charging times and more seamless user experiences. The next steps in this exploration will delve deeper into the logical connection of PLC modems, examining how the correct EV and EVSE modems

establish a secure and reliable communication link before any messages are exchanged. This foundational understanding of high-level communication is key to unlocking the full potential of smart charging and grid integration, paving the way for a truly interconnected and sustainable electric mobility ecosystem.